

Hyunwuk Lee

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thecosmiclab.github.io | hyunwuklee0519.github.io | Google Scholar
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Education

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| Yonsei University, Korea
Integrated Master and Ph.D. in School of Electrical and Electronic Engineering | Mar. 2018 – Aug. 2024 |
| <ul style="list-style-type: none">• GPA: 4.25/4.5• Advisor: Prof. Won Woo Ro• Dissertation: Quantization and Accelerator Designs for Efficient Complex Valued Neural Networks | |
| Yonsei University, Korea
B.S. in School of Electrical and Electronic Engineering | Mar. 2014 – Feb. 2018 |
| <ul style="list-style-type: none">• GPA: 3.61/4.5• Advisor: Prof. Won Woo Ro | |

Experience

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| Assistant Professor
Graduate School of Advanced Semiconductor Engineering, Gwangju Institute of Science and Technology (GIST), Korea | Sep. 2026 – Present |
| <ul style="list-style-type: none">• Director, Computer System and Microarchitecture Design Laboratory (COSMIC Lab) | |
| System Engineer
Samsung Electronics, Hwasung, Korea | Sep. 2024 – Aug. 2026 |
| <ul style="list-style-type: none">• System Architect in SSD FTL Hardware Automation Solution• Modeling, profiling, and designing hardware automation for FTL | |

Research Interest

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| Hardware-Software Co-Design for AI |
| <ul style="list-style-type: none">• Domain-specific accelerators and hardware-aware algorithms• Neural network compression (quantization, sparsity/pruning) and datapath architectures |
| On-Device AI Acceleration |
| <ul style="list-style-type: none">• Edge-efficient execution of multimodal AI (VLA, LLM, diffusion models, etc.)• Profiling latency, memory footprint, and power on real edge hardware |
| System Architecture for AI |
| <ul style="list-style-type: none">• Processing-Near-Memory (PNM), CXL, and KV cache optimization• Multi-GPU systems and data orchestration across nodes |
| Hardware Microarchitecture |
| <ul style="list-style-type: none">• Bandwidth-efficient memory hierarchies and specialized execution units• GPU register and memory systems, cycle-accurate simulation |

Publications

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| SLGS: A Structure-Aware Scanline Renderer for Efficient 3D Gaussian Splatting (To appear) | Oct. 2026 |
| Seunghyun Lee, Mingyu Jung, Sungwoo Kim, Sunghun Kim, Dongho Ha, <i>Hyunwuk Lee</i> , Yingyan (Celine) Lin, and Won Woo Ro | |

- The 35th International Conference on Parallel Architectures and Compilation Techniques (PACT 2026)
- MXFFP: Microscaling Flexible Floating Point Format for Large-Scale AI Model Acceleration** Jun. 2026
 Sungwoo Kim*, Sungbin Kim*, Dongho Ha, *Hyunwuk Lee*, Junsung Kim, Seunghyun Lee, Mingu Jung, Murali Annavaram, and Won Woo Ro
 The 53rd Annual International Symposium on Computer Architecture (ISCA 2026)
 * Co-First Author
- SLICE: A Selective Local Inference Framework with Codec Exploitation for Accelerating Video Super-Resolution** Jun. 2026
 Mingu Jung, Sungbin Kim, Seunghyun Lee, Seunghyun Jin, *Hyunwuk Lee*, and Won Woo Ro
 The 53rd Annual International Symposium on Computer Architecture (ISCA 2026)
Samsung Industry-Academic Research Project Paper Award
- DeSpa: Heterogeneous multi-core accelerators for energy-efficient dense and sparse computation at the tile level in Deep Neural Networks** Mar. 2026
 Hyungjun Jang, Dongho Ha, *Hyunwuk Lee*, and Won Woo Ro
 Journal of Systems Architecture, Vol. 172 (2026)
- BitL: A Hybrid Bit-Serial and Parallel Deep Learning Accelerator for Critical Path Reduction** Oct. 2025
 Seunghyun Lee, Dongho Ha, Sungbin Kim, Sungwoo Kim, *Hyunwuk Lee*, and Won Woo Ro
 The 58th IEEE/ACM International Symposium on Microarchitecture (MICRO 2025)
- Adversarial Purification via Super-Resolution and Diffusion** Oct. 2025
 Mincheol Park, Cheonjun Park, Seungseop Lim, Mijin Koo, *Hyunwuk Lee*, Won Woo Ro, and Suhyun Kim
 IEEE/CVF International Conference on Computer Vision (ICCV 2025)
- CVMAX: Accelerator Architecture with Polar Form Multiplication for Complex-Valued Neural Networks** Jun. 2025
Hyunwuk Lee, Sungbin Kim, Sungwoo Kim, and Won Woo Ro
 The 62nd Design Automation Conference (DAC 2025)
Samsung Industry-Academic Research Project Paper Award
- COSMOS: An LLC Contention Slowdown Model for Heterogeneous Multi-Core Systems** May 2025
 Yongju Lee, Jaewon Kwon, Cheolhwan Kim, Enhyeok Jang, Jiwon Lee, *Hyunwuk Lee*, and Won Woo Ro
 2025 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS 2025)
- Ditto: Accelerating Diffusion Model via Temporal Value Similarity** Mar. 2025
 Sungbin Kim*, *Hyunwuk Lee**, Wonho Cho, Mincheol Park, and Won Woo Ro
 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA 2025)
 * Co-First Author
- REC: Enhancing fine-grained cache coherence protocol in multi-GPU systems** Mar. 2025
 Gun Ko, Jiwon Lee, Hongju Kal, *Hyunwuk Lee*, and Won Woo Ro
 Journal of Systems Architecture, Vol. 160 (2025)
- AirGun: Adaptive Granularity Quantization for Accelerating Large Language Models** Nov. 2024
 Sungbin Kim, *Hyunwuk Lee*, Sungwoo Kim, Cheolhwan Kim, and Won Woo Ro
 The 42nd IEEE International Conference on Computer Design (ICCD 2024)

- GUMSO: Gating Unnecessary On-Chip Memory Slices for Power Optimization on GPUs** Aug. 2024
Seunghyun Jin, *Hyunwuk Lee*, and Won Woo Ro
ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED 2024)
- SHREG: Mitigating register redundancy in GPUs** Jul. 2024
Seunghyun Jin, *Hyunwuk Lee*, Jonghyun Lee, Junsung Kim, and Won Woo Ro
Journal of Systems Architecture, Vol. 152 (2024)
- Exploiting Inherent Properties of Complex Numbers for Accelerating Complex Valued Neural Networks** Oct. 2023
Hyunwuk Lee, Hyungjun Jang, Sungbin Kim, Sungwoo Kim, Wonho Cho, and Won Woo Ro
The 56th International Symposium on Microarchitecture (MICRO 2023)
- Early-Adaptor: An Adaptive Framework for Proactive UVM Memory Management** Apr. 2023
Seokjin Go, *Hyunwuk Lee*, Junsung Kim, Jiwon Lee, Myung Kuk Yoon, and Won Woo Ro
2023 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS 2023)
- Deep Learning with GPUs** Jan. 2021
Won Jeon, Gun Ko, Jiwon Lee, *Hyunwuk Lee*, Dongho Ha, and Won Woo Ro
Book Chapter, Advances in Computers, Vol. 122 (2021)

Patents

- Memory device including a plurality of areas having different refresh periods, memory controller controlling the same and memory system including the same**
Hyunwuk Lee, Gun Ko, Ipoom Jeong, Min Seong Kim, Yong Tag Song, Sung Jae Lee, and Won Woo Ro
US Patent No. 11276452, Korea Application No. 10-2020-0045023, China Patent No. ZL 2020 1 1090146.1
- A method for neural network processing including memory-optimization techniques**
Hongju Kal, Cheonjun Park, *Hyunwuk Lee*, Ipoom Jeong, Jiwon Lee, and Won Woo Ro
Korea Application No. 10-2022-0041848
- Apparatus and method with register sharing**
Seunghyun Jin, Jonghyun Lee, *Hyunwuk Lee*, and Won Woo Ro
US Patent No. 12298901, Korea Application No. 10-2022-0074653
- Memory management unit and method of walking page table**
Jiwon Lee, Ipoom Jeong, Hongju Kal, Gun Ko, *Hyunwuk Lee*, and Won Woo Ro
US Patent No. 12450164, Korea Application No. 10-2022-0175909, China Application No. 202311700848.0
- Memory management apparatus and method for UVM**
Seokjin Go, Junsung Kim, *Hyunwuk Lee*, Jiwon Lee, and Won Woo Ro
Korea Application No. 10-2023-0076606
- Polar Form Aware Apparatus and Method for Complex Valued Neural Network and Rectangular Form Conversion Apparatus**
Hyunwuk Lee, Hyungjun Jang, Sungbin Kim, Wonho Cho, Sungwoo Kim, and Won Woo Ro
Korea Application No. 10-2023-0158538

Honor and Awards

Samsung Industry-Academic Research Project Paper Award Samsung Electronics, Korea — SLICE (ISCA 2026)	2026
Samsung Industry-Academic Research Project Paper Award Samsung Electronics, Korea — CVMAX (DAC 2025)	2025
SK Hynix Industry-Academic Research Project Outstanding Invention SK Hynix, Korea — US Patent No. 11276452	2021

Teaching Experience

Embedded AI System (Graduate) Instructor Graduate School of Advanced Semiconductor Engineering, GIST	Fall 2026
Computer Architecture Teaching Assistant School of Electrical and Electronic Engineering, Yonsei University	2018